Quantum Mechanics 1

Worked Solutions

BPhO Computational Physics Challenge 2026

Introduction

This document gives worked solutions to the *Quantum Mechanics 1* problem sheet. The calculations follow the quantities, conventions and methods used in the supplied BPhO problem sheet and have been checked against the accompanying BPhO solutions. The aim is to show the mathematical reasoning rather than merely state the numerical answers.

1 Question 1

1.1 (i) Atomic and nuclear scales

A typical atomic radius is of order

$$r_{\text{atom}} \sim 10^{-10}\,\text{m},$$

whereas the nuclear radius is of order

$$r_{\text{nucleus}} \sim 10^{-15}\,\text{m}.$$

Hence

$$\frac{r_{\text{atom}}}{r_{\text{nucleus}}} \sim 10^5.$$

Thus the atom is roughly

$$\boxed{10^5}$$

times larger in radius than its nucleus. This illustrates that atoms are mostly empty space, with the positive charge and most of the mass concentrated in the nucleus.

1.2 (ii) Marble and Earth-sized container

The marble diameter is

$$d_m = 3.6\,\text{cm} = 3.6 \times 10^{-2}\,\text{m}.$$

Taking a characteristic atomic diameter of approximately

$$d_a \sim 10^{-10}\,\text{m},$$

the number of atoms across a linear dimension of the marble is approximately d_m/d_a, so the number of atoms is of order

$$N_{\text{atom}} \sim \left(\frac{3.6 \times 10^{-2}}{10^{-10}}\right)^3 \approx 4.7 \times 10^{25}.$$

Therefore

$$\boxed{N_{\text{atom}} \sim 4.7 \times 10^{25}}.$$

Now take the Earth diameter to be

$$d_E = 1.28 \times 10^7\,\mathrm{m}.$$

Ignoring the space between marbles, the number of marbles that would fill an Earth-sized volume is

$$N_{\text{marble}} \sim \left(\frac{1.28 \times 10^7}{3.6 \times 10^{-2}}\right)^3 \approx 4.5 \times 10^{25}.$$

Hence

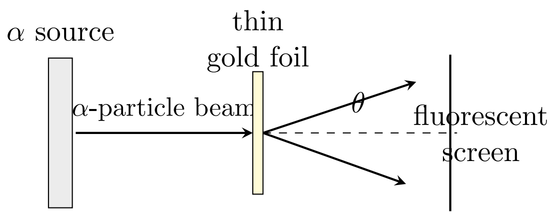
$$\boxed{N_{\text{marble}} \sim 4.5 \times 10^{25}}.$$

The striking result is that these numbers are of the same order of magnitude: the number of atoms in a large marble is comparable to the number of marbles needed to fill an Earth-sized volume.

2 Question 2 — Rutherford scattering

The Rutherford experiment was performed by Geiger and Marsden under Rutherford's direction between 1908 and 1913. A beam of α particles was directed at a very thin sheet of gold foil, and the scattered particles were detected using a fluorescent screen.

A schematic representation of the apparatus is



Most α particles passed through the foil essentially undeflected. Some were deflected through appreciable angles, and a very small number were scattered through very large angles, including particles that were effectively "backscattered."

The α particle is positively charged, so the large-angle deflections indicate strong electrostatic repulsion from a concentrated positive charge. The conclusions were therefore:

- atoms are mostly empty space;
- the positive charge is concentrated in a very small region;
- almost all of the atomic mass is concentrated in this region;
- this region is the atomic nucleus, with a characteristic scale of about 10^{-15} m compared with an atomic scale of about 10^{-10} m.

The experiment therefore provided decisive evidence for the nuclear model of the atom.

3 Question 3 — Thermal radiation

The Stefan–Boltzmann law is

$$\Phi = \sigma T^4,$$

where

$$\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}.$$

For a radiating area A,

$$P = A\sigma T^4.$$

Wien's displacement law is

$$\lambda_{\text{max}} T = 2.90 \times 10^{-3}\,\text{m K}.$$

3.1 (i) Temperature of a tungsten filament

The filament is modelled as a thin strip with

$$L = 0.20\,\text{m}, \qquad w = 0.14\,\text{mm} = 1.4 \times 10^{-4}\,\text{m}.$$

Using the strip area specified by the model,

$$A = Lw = (0.20)(1.4 \times 10^{-4}) = 2.8 \times 10^{-5}\,\text{m}^2.$$

We make the simplified model assumption that the effective radiating area is the given strip area $A = Lw$ and that the full 100 W electrical input is emitted as thermal radiation. Thus

$$P = A\sigma T^4,$$

so

$$T = \left(\frac{P}{A\sigma}\right)^{1/4}.$$

Therefore

$$T = \left[\frac{100}{(2.8 \times 10^{-5})(5.67 \times 10^{-8})}\right]^{1/4} \approx 2820\,\text{K}.$$

Converting to Celsius,

$$T \approx 2820 - 273 \approx 2540^\circ\text{C}.$$

Thus

$$\boxed{T \approx 2540^\circ\text{C}}.$$

The assumptions are that the effective radiating area is Lw, that the full 100 W electrical input is radiated thermally, and that the filament behaves as a black body. These are approximations: a real bulb also loses energy through conduction and convection, and tungsten is not a perfect black body. The resulting temperature is therefore an estimate, but is reasonable for the simplified model.

3.2 (ii) Luminosity of the Sun

The solar radius is

$$R_\odot = 695500\,\text{km} = 6.955 \times 10^8\,\text{m},$$

and its surface temperature is

$$T_\odot = 5778\,\text{K}.$$

The surface area is

$$A_\odot = 4\pi R_\odot^2.$$

Therefore

$$L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4.$$

Substitution gives

$$L_\odot = 4\pi (6.955 \times 10^8)^2 (5.67 \times 10^{-8})(5778)^4,$$

hence

$$\boxed{L_\odot \approx 3.84 \times 10^{26}\,\text{W}}.$$

3.3 (iii) Solar panel on Mars

At distance

$$r = 227.9\,\text{million km} = 2.279 \times 10^{11}\,\text{m},$$

the Sun's radiation is spread over a sphere of area

$$4\pi r^2.$$

A square panel of area a^2 therefore receives

$$P_{\text{received}} = \frac{L_\odot}{4\pi r^2} a^2.$$

If the panel efficiency is $\beta = 0.20$, its useful output is

$$P = \beta \frac{L_\odot}{4\pi r^2} a^2.$$

Solving for the side length,

$$a = \sqrt{\frac{4\pi P r^2}{\beta L_\odot}}.$$

For $P = 100\,\text{W}$,

$$a = \sqrt{\frac{4\pi(100)(2.279 \times 10^{11})^2}{(0.20)(3.84 \times 10^{26})}} \approx 0.92\,\text{m}.$$

Thus

$$\boxed{a \approx 0.92\,\text{m}}.$$

3.4 (iv) Radiation from human skin

Taking the skin temperature as

$$T = 34^\circ\text{C} = 307\,\text{K},$$

Wien's law gives

$$\lambda_{\text{max}} = \frac{2.90 \times 10^{-3}}{307}.$$

Hence

$$\lambda_{\text{max}} \approx 9.45 \times 10^{-6}\,\text{m} = \boxed{9450\,\text{nm}}.$$

This lies in the

$$\boxed{\text{infrared}}$$

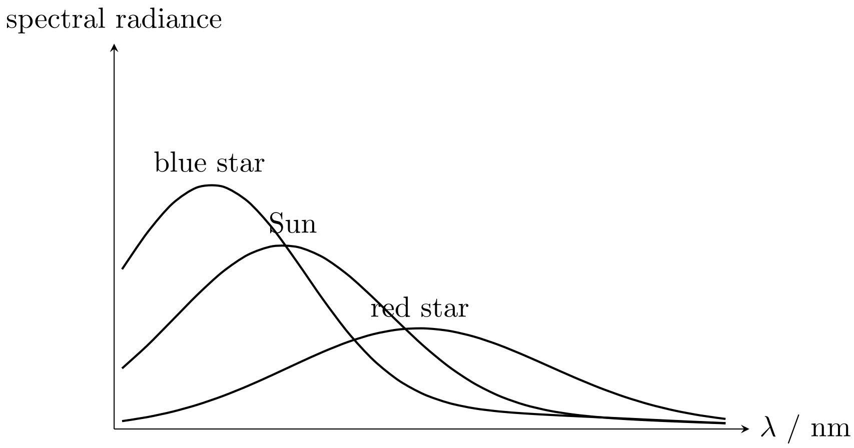
region of the electromagnetic spectrum, explaining the use of infrared cameras for detecting heat signatures.

3.5 (v) Spectra of hot and cool stars

The Planck spectrum has the form

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_B T)} - 1}.$$

A qualitative sketch is shown below. The hotter blue star peaks at a shorter wavelength and has a larger total emitted power, while the cooler red star peaks at a longer wavelength. The total area under each curve increases with temperature because of the T^4 dependence of the Stefan–Boltzmann law.



For the Sun,

$$\lambda_{\text{max}} = \frac{2.90 \times 10^{-3}}{5778} \approx 5.02 \times 10^{-7}\,\text{m},$$

so

$$\boxed{\lambda_{\text{max}} \approx 502\,\text{nm}}.$$

3.6 (vi) Expansion into a red giant

Initially,

$$\lambda_{\text{max},\odot} \approx 502\,\text{nm}.$$

For the red giant,

$$\lambda_{\text{max,RG}} = 680\,\text{nm}.$$

Using Wien's law,

$$T_{\text{RG}} = \frac{2.90 \times 10^{-3}}{680 \times 10^{-9}} \approx 4265\,\text{K}.$$

Assuming the luminosity remains constant,

$$L = 4\pi R^2 \sigma T^4 = \text{constant}.$$

Therefore

$$R_{\text{RG}}^2 T_{\text{RG}}^4 = R_\odot^2 T_\odot^4,$$

giving

$$\frac{R_{\text{RG}}}{R_\odot} = \left(\frac{T_\odot}{T_{\text{RG}}}\right)^2.$$

Thus

$$\frac{R_{\text{RG}}}{R_\odot} = \left(\frac{5778}{4265}\right)^2 \approx 1.84.$$

Hence

$$\boxed{\frac{R_{\text{RG}}}{R_\odot} \approx 1.84}.$$

4 Question 4 — Photoelectric effect

For silver,

$$\phi = 4.3\,\text{eV}.$$

The photoelectric equation is

$$K_{\text{max}} = hf - \phi = \frac{hc}{\lambda} - \phi.$$

4.1 (i) Maximum wavelength

At threshold,

$$K_{\text{max}} = 0,$$

so

$$\frac{hc}{\lambda_{\text{max}}} = \phi.$$

Hence

$$\lambda_{\text{max}} = \frac{hc}{\phi}.$$

Using $\phi = 4.3 \times 1.60 \times 10^{-19}\,\text{J}$,

$$\lambda_{\text{max}} = \frac{(6.63 \times 10^{-34})(2.998 \times 10^{8})}{(4.3)(1.60 \times 10^{-19})} \approx 2.89 \times 10^{-7}\,\text{m}.$$

Therefore

$$\boxed{\lambda_{\text{max}} \approx 289\,\text{nm}}.$$

4.2 (ii) Photoelectron energy for 200 nm light

The photon energy is

$$E_\gamma = \frac{hc}{\lambda}.$$

Thus

$$K_{\text{max}} = \frac{hc}{\lambda} - \phi.$$

For $\lambda = 200\,\text{nm}$,

$$K_{\text{max}} \approx 1.91\,\text{eV}.$$

Converting to joules,

$$K_{\text{max}} = (1.91)(1.60 \times 10^{-19}) \approx 3.06 \times 10^{-19}\,\text{J}.$$

Hence

$$\boxed{K_{\text{max}} \approx 3.06 \times 10^{-19}\,\text{J}}.$$

4.3 (iii) Photon absorption rate

The incident power is

$$P = 1.00 \times 10^{-6}\,\text{W}.$$

Each photon has energy

$$E_\gamma = \frac{hc}{\lambda}.$$

Therefore the number absorbed per second is

$$\dot{N} = \frac{P}{E_\gamma} = \frac{P\lambda}{hc}.$$

Thus

$$\dot{N} \approx 1.01 \times 10^{12}\,\text{s}^{-1}.$$

So

$$\boxed{\dot{N} \approx 1.01 \times 10^{12}\,\text{photons}\,\text{s}^{-1}}.$$

4.4 (iv) Photocurrent

If one electron is emitted per photon,

$$I = e\dot{N}.$$

Hence

$$I = (1.60 \times 10^{-19})(1.01 \times 10^{12}) \approx 1.61 \times 10^{-7}\,\mathrm{A}.$$

Therefore

$$\boxed{I \approx 1.61 \times 10^{-7}\,\mathrm{A}}$$

or approximately

$$\boxed{0.16\,\mu\mathrm{A}}.$$

4.5 (v) Stopping voltage

The stopping voltage satisfies

$$eV_s = K_{\max}.$$

Since $K_{\max} = 1.91\,\mathrm{eV}$,

$$\boxed{V_s \approx 1.91\,\mathrm{V}}.$$

5 Question 5 — Electron diffraction

An electron of mass m and charge magnitude e is accelerated through a potential difference V.

5.1 (i) Final velocity

The electron gains kinetic energy

$$eV.$$

In the classical limit,

$$eV = \frac{1}{2}mv^2.$$

Therefore

$$v^2 = \frac{2eV}{m},$$

and

$$\boxed{v = \sqrt{\frac{2eV}{m}}}.$$

5.2 (ii) Classical limit

We require

$$v < 0.01c.$$

Using the result above,

$$\sqrt{\frac{2eV}{m}} < 0.01c.$$

Squaring,

$$\frac{2eV}{m} < 10^{-4}c^2,$$

so

$$V < 10^{-4}\frac{mc^2}{2e}.$$

For an electron,

$$m = 9.109 \times 10^{-31}\,\mathrm{kg}, \qquad e = 1.60 \times 10^{-19}\,\mathrm{C},$$

and

$$c = 2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}.$$

Therefore

$$\boxed{V < 25.6\,\mathrm{V}}.$$

Below this voltage, relativistic effects can safely be neglected.

For comparison, if one extrapolates the classical expression to very high voltages, the discrepancy eventually becomes important. The exact relativistic kinetic energy is

$$eV = (\gamma - 1)mc^2, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

For example, at $V = 5\,\mathrm{kV}$ the classical approximation gives $v/c \approx 0.14$, so relativistic corrections are no longer completely negligible.

5.3 (iii) Classical de Broglie wavelength

The de Broglie relation is

$$\lambda = \frac{h}{p}.$$

In the classical limit,

$$K = \frac{p^2}{2m}.$$

Since

$$K = eV,$$

we have

$$eV = \frac{p^2}{2m}.$$

Therefore

$$p = \sqrt{2meV},$$

and hence

$$\boxed{\lambda = \frac{h}{\sqrt{2meV}}}.$$

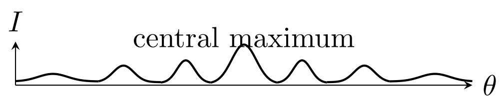
5.4 (iv) Diffraction pattern

The electron beam strikes a carbon disc whose atomic layers have spacing

$$d = 1.23 \times 10^{-10}\,\mathrm{m}.$$

The screen therefore shows a diffraction/interference pattern rather than a single classical scattering spot.

A qualitative sketch of intensity against scattering angle is



The maxima arise because the electron has a de Broglie wavelength

$$\lambda = \frac{h}{p}$$

and waves scattered from successive atomic layers interfere constructively at particular angles.

Thus the experiment demonstrates that electrons possess wave-like properties: they can diffract and interfere.

5.5 (v) First diffraction peak

For the geometry used in the problem sheet, the first-order condition is

$$d \sin\theta = \lambda.$$

For the first peak,

$$\theta = 10.0^\circ, \qquad d = 1.23 \times 10^{-10}\,\mathrm{m}.$$

Using

$$\lambda = \frac{h}{\sqrt{2meV}},$$

we obtain

$$d \sin\theta = \frac{h}{\sqrt{2meV}}.$$

Squaring and rearranging,

$$V = \frac{h^2}{2me\, d^2 \sin^2\theta}.$$

Therefore

$$V = \frac{(6.63 \times 10^{-34})^2}{2(9.109 \times 10^{-31})(1.60 \times 10^{-19})(1.23 \times 10^{-10})^2 \sin^2(10^\circ)} \approx 3.3 \times 10^3\,\mathrm{V}.$$

Hence

$$\boxed{V \approx 3.3\,\mathrm{kV}}.$$

5.6 (vi) Other diffraction peaks

For diffraction order n,

$$d \sin\theta_n = n\lambda.$$

Thus

$$\boxed{\theta_n = \sin^{-1}\left(\frac{n\lambda}{d}\right)}.$$

For the voltage calculated above,

$$\frac{\lambda}{d} \approx 0.174.$$

Hence

$$\theta_n = \sin^{-1}(0.174n).$$

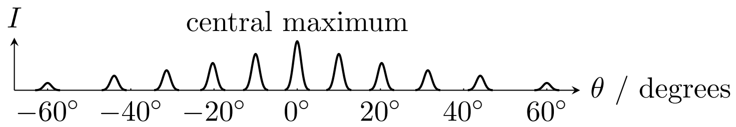
The allowed positive orders are $n = 1, \ldots, 5$, since

$$\frac{1}{0.174} \approx 5.76.$$

The corresponding scattering angles are approximately

$$\boxed{\theta = 0^\circ,\ \pm 10.0^\circ,\ \pm 20.3^\circ,\ \pm 31.4^\circ,\ \pm 44.0^\circ,\ \pm 60.3^\circ}.$$

The central maximum occurs at 0°, with symmetric maxima on either side. A qualitative sketch of the resulting pattern is



Because the carbon disc and detector have finite dimensions, the real peaks have finite widths rather than being infinitely sharp.

6 Question 6 — Hydrogen spectral lines

The Bohr model gives the hydrogen energy levels as

$$\boxed{E_n = -\frac{13.6\,\mathrm{eV}}{n^2}}.$$

For a transition from n to $m < n$, the emitted photon has energy

$$E_\gamma = E_n - E_m$$

in magnitude, and

$$E_\gamma = \frac{hc}{\lambda}.$$

Equivalently, the wavelength is given by the supplied hydrogenic relation

$$\lambda_{nm} \approx 91.13\,\mathrm{nm}\left(\frac{1}{m^2} - \frac{1}{n^2}\right)^{-1}.$$

6.1 (i) Lyman series: $n = 3 \to 1$

For $m = 1$ and $n = 3$,

$$\lambda = \frac{91.13\,\mathrm{nm}}{1 - \frac{1}{3^2}}.$$

Therefore

$$\lambda \approx 103\,\mathrm{nm}.$$

Thus

$$\boxed{\lambda \approx 103\,\mathrm{nm}},$$

which lies in the

$$\boxed{\text{ultraviolet}}$$

region.

6.2 (ii) Balmer series: $n = 4 \to 2$

Now

$$\lambda = \frac{91.13\,\mathrm{nm}}{\frac{1}{2^2} - \frac{1}{4^2}}.$$

Hence

$$\lambda \approx 487\,\mathrm{nm}.$$

Therefore

$$\boxed{\lambda \approx 487\,\mathrm{nm}},$$

which lies in the

$$\boxed{\text{visible}}$$

region.

6.3 (iii) Pfund series: $n = 7 \to 5$

For $m = 5$ and $n = 7$,

$$\lambda = \frac{91.13\,\text{nm}}{\frac{1}{5^2} - \frac{1}{7^2}}.$$

Therefore

$$\lambda \approx 4660\,\text{nm}.$$

Hence

$$\boxed{\lambda \approx 4660\,\text{nm}},$$

which lies in the

$$\boxed{\text{infrared}}$$

region.

Numerical results

Part	Result
1(i)	$r_{\text{atom}}/r_{\text{nucleus}} \sim 10^5$
1(ii)	$N_{\text{atom}} \sim 4.7 \times 10^{25}$; $N_{\text{marble}} \sim 4.5 \times 10^{25}$
3(i)	$T \approx 2540°\text{C}$
3(ii)	$L_\odot \approx 3.84 \times 10^{26}\,\text{W}$
3(iii)	$a \approx 0.92\,\text{m}$
3(iv)	$\lambda_{\max} \approx 9450\,\text{nm}$ (IR)
3(v)	$\lambda_{\max,\odot} \approx 502\,\text{nm}$
3(vi)	$R_{\text{RG}}/R_\odot \approx 1.84$
4(i)	$\lambda_{\max} \approx 289\,\text{nm}$
4(ii)	$K_{\max} \approx 3.06 \times 10^{-19}\,\text{J}$
4(iii)	$\dot{N} \approx 1.01 \times 10^{12}\,\text{s}^{-1}$
4(iv)	$I \approx 1.61 \times 10^{-7}\,\text{A}$
4(v)	$V_s \approx 1.91\,\text{V}$
5(i)	$v = \sqrt{2eV/m}$
5(ii)	$V < 25.6\,\text{V}$
5(iii)	$\lambda = h/\sqrt{2meV}$
5(iv)	diffraction/interference pattern; electrons exhibit wave-like behaviour
5(v)	$V \approx 3.3\,\text{kV}$
5(vi)	$\theta = 0°, \pm 10.0°, \pm 20.3°, \pm 31.4°, \pm 44.0°, \pm 60.3°$
6(i)	103 nm (UV)
6(ii)	487 nm (visible)
6(iii)	4660 nm (IR)